

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CE/IT)

April/May 2012

CE-201 Data Structure and Algorithms

Date: 01.05.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Answer the following questions.

[07]

(a) What do you mean by complexity of an algorithm? Explain the meaning of worst case analysis and best case analysis.

[03]

(b) Derive the equation for computing the address of the memory location of an element stored as row major order in a two-dimensional array.

[02]

(c) How does doubly linked list differ from singly linked list?

[02]

Q - 2 Answer the following questions.

[14]

(a) What do you mean by hash clash? Explain in detail any one method to resolve hash collisions.

[04]

(b) Define Sorting. Write an algorithm for Selection sort. Apply the same algorithm to sort the elements 25 15 30 9 99 20 26 80

[07]

(c) Assume $A = 1$, $B = 2$, $C = 3$. Evaluate the following postfix expression using stack:

[03]

$A B C + * C B A - + *$

OR

Q - 2 Answer the following questions.

[14]

(a) Convert the following Infix expression to Postfix form using a stack:

[04]

$x + y * z + (p * q + r) * s$

(b) Sort the following numbers using merge sort.

[05]

42 23 74 11 65 57 94 36 99 87

(c) Using the hash function $h(i) = (2i + 5) \bmod 11$, represent the chained hash table.

[05]

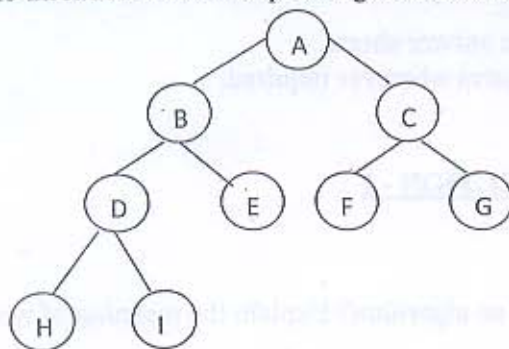
Consider the set of key {12, 44, 13, 88, 23, 94, 11, 39, 20, 16, 5}

Q – 3 Answer the following questions. [14]

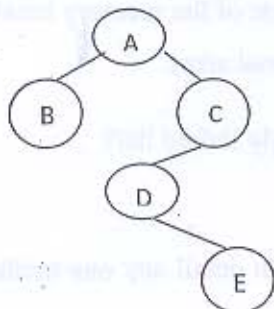
(a) Write an algorithm to insert a node in between any two nodes in a singly linked list. [04]

(b) In a circular queue show the conditions that differentiate between empty queue and full queue. Also explain disadvantages of implementing queue (non circular) using array. [04]

(c) Draw the threaded tree corresponding to the following binary tree. [03]



(d) Determine whether the following binary tree is an AVL tree? Give reasons for your claim. [03]



OR

Q – 3 Answer the following questions. [14]

(a) Write a C program using function which deletes the last element of a singly linked list. [04]

(b) Define Priority Queues. Explain how priority queues can be implemented. [04]

(c) Write an algorithm for finding solution to the Towers of Hanoi problem. Explain the working of your algorithm (4 disks) with diagrams. [06]

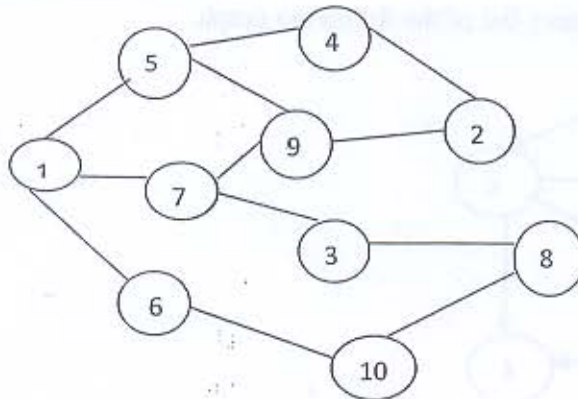
SECTION – II

Q – 4 Write an algorithm for Binary Search technique. [07]

Apply the algorithm on an ordered array A with the following elements {11, 22, 30, 33, 40, 44, 55, 60, 66, 77, 80, 88, 99}. Determine the number of key comparisons made while searching for keys 40 and 85.

Q - 5 Answer the following questions. [14]

- (a) What is a height balanced tree? Explain how the height is balanced after addition of nodes in it? [08]
- (b) Consider the undirected graph G given below. Apply depth-first search traversal starting at vertex 1. [04]



- (c) Explain double ended queue. [02]

OR

Q - 5 Answer the following questions. [14]

- (a) What is a Binary Search Tree (BST)? Make a BST for following sequence. [08]
416, 891, 456, 765, 111, 654, 345, 256, 333.

Perform following operations on BST.

- 1) Insert a key 332 2) Delete a key 891

- (b) Write down any four applications of a graph. [02]

- (c) Write a short note on sequential file organization. [04]

Q - 6 Answer the following questions. [14]

- (a) Consider a queue whose capacity is 6. Following operations are carried out in the sequence shown. Initially rear=front=0 [05]

- (i) Objects A, B, D, H added
- (ii) An object removed
- (iii) Object I added
- (iv) An object removed
- (v) Object C, F, G, K added

Show the 5 queue representations with rear and front pointer after each operation is carried out.

- (b) Define a connected graph. For a connected graph of n vertices, how many [05]

maximum and minimum edges should be in the graph? Prove it with example.

(c) Differentiate BFS and DFS.

[02]

(d) Define a stack. Describe ways to implement stack.

[02]

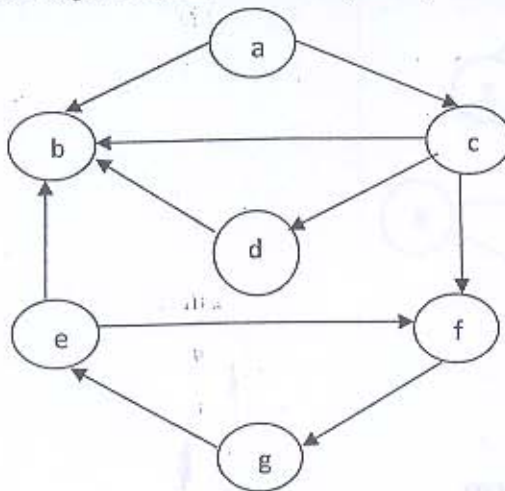
OR

Q - 6 Answer the following questions.

[14]

(a) Give the adjacency matrix and adjacency list of the following graph.

[04]



(b) Define a sparse matrix. Explain different types of sparse matrices?

[04]

(c) Define a linked-list. How are these stored in the memory? Write a procedure to find the maximum of the values in the list. Assume the linked list in the memory consisting of numerical data values.

[06]
